

Lower-Body Biomechanics Pipeline for Kinematic and Kinetic Analysis from Raw Marker Data

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This report describes a custom processing chain implemented in Python for inverse kinematics and inverse dynamics of walking data. Section labels follow the course submission checklist.

1 Introduction

Human movement analysis sits at the intersection of clinical decision making, rehabilitation engineering, and basic science. For the lower body in particular, clinicians and researchers routinely ask quantitative questions that are difficult to answer from observation alone. How much load does a joint experience during stance? How does coordination change after surgery? Does an intervention alter the timing or magnitude of knee flexion during weight acceptance? Optical motion capture, synchronized with force platforms, remains a practical gold standard for answering these questions in controlled laboratory settings because it provides dense kinematic records alongside direct measurement of external loads.

Turning raw laboratory signals into interpretable biomechanical variables is not a single calculation but a chain of interdependent modeling decisions. Choices regarding coordinate systems, rigid body assumptions, filtering, segment inertial properties, and inverse problem formulation all influence the final outputs. Small inconsistencies, such as an anatomical axis that is correct in isolation but misaligned relative to another segment, or a center of pressure definition that does not match the reference frame used for moments, can produce joint angles and moments that appear plausible yet fail basic physical checks. If left unrecognized, these errors can lead to incorrect interpretations of joint loading or movement strategy and may undermine both clinical insight and research conclusions.

For student projects and translational work, there is an additional requirement that workflows remain transparent and adaptable. Future users should be able to trace a reported knee moment back through filtering choices, rigid body kinematics, and the external wrench applied at the foot. This is particularly important in applications such as gait assessment and rehabilitation monitoring, where subject specific variability and evolving hypotheses are common. In these settings, rapid iteration, adjustable parameters, and explicit assumptions are often as important as achieving high accuracy in a single analysis pass.

The work summarized here develops and documents a modular biomechanical analysis pipeline that begins with three dimensional marker trajectories stored in C3D format and

produces time series of joint angles and net joint moments for the pelvis to foot lower limb chain. The implementation is intentionally structured as a transparent and modular system rather than a monolithic black box tool. Each major stage is implemented in an independent script, and parameters are exposed at stage boundaries. Intermediate artifacts, such as CSV joint center outputs and NPZ kinematic exports, enable selective recomputation and support debugging, validation, and iterative refinement without requiring full pipeline re-execution.

The remainder of the report mirrors the computational flow and the design process. A concise prior-art section situates the work in biomechanics literature without implementation detail. A formal needs statement and design goals justify engineering priorities. A methods section states the implemented equations by module. Testing methods and iterative testing results explain how evaluation was structured and how failures traced to specific assumptions tracking versus anatomical segment frames, knee joint coordinates, center-of-pressure handling, and Newton–Euler integration producing the conventions in the final code. Representative inverse kinematics and inverse dynamics outputs from a walking trial appear in the results section. A short section records poster feedback and follow-up work. The discussion assesses what the design accomplishes, what remains fragile, and sensible next steps.

The engineering scope is deliberately bounded. The pipeline estimates rigid body kinematics and classical inverse dynamics for the lower limb chain. It does not estimate muscle forces, joint contact pressures, or patient specific ligament contributions. These omissions are intentional. They reflect a design decision to first establish a reliable kinematic and kinetic foundation that can later be extended with musculoskeletal optimization or machine learning approaches if needed. In this sense, the project is best understood as infrastructure. It is the part of the workflow that should be unremarkable when correct and immediately obvious when something is wrong.

2 Background and Prior Art

2.1 Motion capture and rigid segment kinematics

Retroreflective markers on the skin move relative to bone, so cluster-based rigid tracking is always approximate yet remains standard for gait. Least squares rigid alignment between corresponding template and moving points (Kabsch and equivalent SVD formulations) supplies segment orientation and translation [3].

2.2 Joint conventions

ISB-style recommendations reduce arbitrary variation in lower-body axis definitions [2]. For the knee, Grood–Suntay separates flexion–extension, abduction–adduction, and internal–external rotation using a femoral flexion axis and floating axes tied to long-bone directions, which behaves more stably than naive Euler sequences near singular configurations [4].

2.3 Kinetics, inertia, and software context

Inverse dynamics estimates intersegmental forces and moments from external loads and rigid-body kinematics [5]. Mass and inertia are often scaled from anthropometric tables when direct measurement is unavailable. The foot wrench combines ground reaction force, center of pressure, and free moment; lever arms from COP to joint centers dominate the ankle solution, so plate metadata, units, and filtering across contact transitions matter as much as the propagation equations [5].

OpenSim illustrates a mature path from laboratory motion to musculoskeletal simulation [1]; smaller inspectable pipelines remain useful when exploratory iteration outweighs full model preparation. Reporting guidance stresses explicit coordinate and sign metadata for intersegmental moments so curves remain interpretable [7]. Despite these established methods, many biomechanics workflows remain difficult to inspect and modify at intermediate stages, particularly in educational and exploratory research settings. This motivates the

development of simpler, transparent pipelines that preserve established conventions while improving accessibility and iterative flexibility.

3 Formal Needs Statement and Design Goals

3.1 Needs statement

OpenSim class tools[1] provide validated musculoskeletal modeling capabilities but require substantial model preparation and workflow overhead. For design exploration, coursework scale replication, pilot studies, and methods prototyping, this overhead often exceeds the immediate needs of the analysis. In these cases, a customizable script based pipeline with explicit assumptions and inspectable intermediate stages offers a practical alternative without attempting to replace full muscle-driven simulation.

3.2 Design goals

The following goals were chosen to be simultaneously measurable and protective against common failure modes.

Consistent anatomical coordinate systems. Angles and moments are only comparable if axis definitions are stable across static calibration, dynamic fitting, and inverse dynamics projection. The pipeline therefore commits to explicit pelvis construction following ISB-style logic in the static calibration stage and propagates handedness checks downstream.

Robust marker-based tracking. Rather than relying on a single heuristic per segment, the implementation uses correspondence-based rigid transforms (Kabsch via SVD) between static and moving marker clusters, with guards for missing markers and for degenerate point sets.

Clinically meaningful joint angles. Clinically interpretable joint angles were defined using conventions consistent with standard laboratory reporting. Hip angles follow the project’s established decomposition, knee kinematics use the Grood–Suntay convention,

and ankle axes are defined consistently with the segment coordinate systems used in inverse dynamics.

Accurate joint moment estimation, with physical interpretability. Inverse dynamics uses a Newton–Euler formulation with the ground reaction wrench applied at the center of pressure, explicit Euler torques from segment angular velocity, and moment projection into the same joint coordinate systems used for kinematics.

Transparent modular design. Each stage is callable and re-runnable; intermediate exports exist so that angle-only replotting can occur without repeating SVD fits. Stages communicate through explicit files (CSV, NPZ), which keeps responsibilities separated and makes it straightforward to swap or extend one component without rewriting the whole pipeline (Section 9.1).

Highly adjustable parameters. Filter cutoffs, stance detection thresholds, and export paths are parameterized rather than hard-coded, acknowledging that acceptable smoothing depends on sampling rate, task, and noise floor.

3.3 Scope boundaries and operational assumptions

Several assumptions are not argued as universal truths; they are engineering commitments chosen to keep the semester project tractable. The pipeline assumes rigid clusters approximate bone motion well enough for slow and moderate-speed walking in a controlled lab. It assumes force-plate data are synchronized to marker data after accounting for acquisition clock offsets and export conventions. It assumes segment inertia can be approximated using Winter-style scaling from measured or reported body mass. It assumes joint centers exported from static calibration are preferable to purely geometric midpoints for some landmarks, while still acknowledging that functional hip joint center estimation from markers alone is an approximation.

The scope boundaries matter for honest interpretation. The pipeline does not estimate ligament forces, cartilage contact pressure, or muscle coordination. It does not perform joint

optimization across a musculoskeletal graph. It does not automatically detect marker label swaps beyond what missing-data guards reveal. Those boundaries are compatible with the needs statement: the objective is a transparent backbone for kinematics and kinetics, not a replacement for every analysis mode available in mature musculoskeletal software.

4 Methods

Notation: the laboratory frame is implicit unless noted. $\mathbf{R} \in SO(3)$ maps a segment body basis into the lab ($\mathbf{v}_{\text{lab}} = \mathbf{R}\mathbf{v}_{\text{body}}$). Scalars m , $\mathbf{r}$, $\boldsymbol{\omega}$, and $\boldsymbol{\alpha}$ denote mass, position, angular velocity, and angular acceleration; $\mathbf{g} = [0, 0, -g]^\top$ with $g \approx 9.81 \text{ m s}^{-2}$. Stages exchange CSV and NPZ files; script names below match the repository.

4.1 Static calibration (`static_calibration.py`)

Each segment obtains a **tracking frame (TF)** from its static marker cluster. The origin is placed at the centroid, and the basis is formed by a right-handed orthonormalization of the dominant directions of centered marker coordinates using a thin SVD. For the pelvis, a geometry based construction from ASIS/PSIS landmarks is used when available.

An **anatomical coordinate system (ACS)** is built from bony landmarks and joint-center definitions (e.g. pelvis from ASIS/PSIS, femur long axis from knee to hip with anterior direction hints, tibia from ankle to knee, and foot defined at the ankle with plantar hinge logic).

The rigid offset from ACS to TF in the static pose, rotation $\mathbf{R}_{TA}$ and translation $\mathbf{t}_{TA}$, is stored along with template marker coordinates $\mathbf{C}_{\text{local}}$ expressed in the TF. During walking, the tracked TF pose ($\mathbf{R}_T^{\text{world}}, \mathbf{o}_T$) composes as

$$\mathbf{R}_A^{\text{world}} = \mathbf{R}_T^{\text{world}} \mathbf{R}_{TA}, \quad \mathbf{o}_A = \mathbf{o}_T + \mathbf{R}_T^{\text{world}} \mathbf{t}_{TA}. \quad (1)$$

Joint centers (including regression-based hip estimates where configured) are exported to

CSV for use in length-based inertia calculations.

4.2 Dynamic rigid fitting (`svd_kabsch.py`)

For each segment and frame, corresponding rows of $\mathbf{C}_{\text{local}}$ (TF) and lab markers $\mathbf{q}_i$ minimize $\sum_i \|\mathbf{q}_i - (\mathbf{R}\mathbf{p}_i + \mathbf{t})\|^2$ over $\mathbf{R} \in SO(3)$ and $\mathbf{t} \in \mathbb{R}^3$. This is solved via centroid removal and SVD of the 3×3 cross-covariance between centered point sets (Kabsch/Umeyama implementation).

Optional uniform scaling is available but typically fixed to unity for standard marker sets. NPZ exports store per-frame $\mathbf{R}_A^{\text{world}}$ and $\mathbf{o}_A$ after composition with (1).

4.3 Joint angles and QC plotting (`angles_only.py`)

Joint angles are computed as signed rotations in planes normal to selected axes using a two-argument arctangent of projected vectors. The knee follows the Grood–Suntay convention [4], using a femoral flexion axis and floating axes derived from femur and tibia ACS bases expressed in the laboratory frame. A discrete $\pm$ branch is resolved by selecting the solution with the smaller internal rotation magnitude.

Hip and ankle angles are obtained from relative rotations between pelvis, femur, tibia, and foot ACS matrices, with documented reflections and fixed $\pi/2$ reorientations where implemented.

`angles_only.py` loads saved kinematics NPZ files, computes joint angles, and generates plots without repeating the rigid body fitting stage. It also supports zero-referenced visualization by subtracting mean hip and knee values from a static trial.

4.4 Force-plate preprocessing (`forceplate_preprocess.py`)

With ground reaction force $\mathbf{F}$ and plate moment $\mathbf{M}_O$ about reference point O , and during stance where $|F_z| > \varepsilon$, planar center of pressure (COP) in plate coordinates is computed

from horizontal moment balance:

$$\text{COP}_x = -\frac{M_y}{F_z}, \quad \text{COP}_y = \frac{M_x}{F_z}. \quad (2)$$

The vertical free moment is transported between O and the COP using $\mathbf{r}_{\text{COP}} \times \mathbf{F}$. Plate geometry metadata (corners and origin) are used to map $\mathbf{r}_{\text{COP}}$ into the laboratory frame. Force, moment, and length units are converted consistently to SI units before use in dynamics.

Filtering is applied only during stance phases to avoid interpolating COP across swing when vertical force indicates no contact.

4.5 Segment inertia (`inertial_segments.py`)

Segment lengths derived from calibrated joint centers are used to scale mass and center-of-mass location along each segment using Winter style anthropometric fractions.

About the COM, each segment is assigned a diagonal inertia tensor. The longitudinal component uses $k = \kappa L$ along the segment axis, and transverse components are approximated using a thin-rod formulation $mL^2/12$ orthogonal to that axis. These tensors are expressed in the segment ACS prior to use in the inverse dynamics solver.

4.6 Kinematic derivatives (`kinematic_derivatives.py`)

Marker-derived positions and rotation matrices are filtered using a zero-phase, fourth order Butterworth filter (`filtfilt`) at documented cutoff frequencies. Linear velocity and acceleration are computed using central differences on filtered positions, with one-sided differences applied at endpoints.

Angular velocity is recovered from successive rotation matrices $\mathbf{R}_{k-1}^T \mathbf{R}_k$ using the skew-symmetric component scaled by $1/(2\Delta t)$. Angular acceleration is obtained from filtered finite differences of $\boldsymbol{\omega}$. Ground reaction signals are filtered on the same time base when

available.

4.7 Inverse dynamics (`inverse_dynamics_newton_euler.py`)

For the foot, with center-of-mass acceleration $\mathbf{a}$, inertia $\mathbf{I}$ about the COM in the foot frame, and orientation $\mathbf{R}_f$, linear momentum balance gives

$$\mathbf{F}_{\text{ankle}} = m\mathbf{a} - \mathbf{F}_{\text{grf}} - m\mathbf{g}. \quad (3)$$

Euler’s equation in the body frame,

$$\boldsymbol{\tau}_b = \mathbf{I}\boldsymbol{\alpha}_b + \boldsymbol{\omega}_b \times (\mathbf{I}\boldsymbol{\omega}_b),$$

is mapped to the laboratory frame as $\boldsymbol{\tau}_{\text{lab}} = \mathbf{R}_f\boldsymbol{\tau}_b$.

The ankle reaction moment is obtained from moment balance about the COM, including the ground reaction force applied at the COP and the ankle reaction force. The solution is then propagated proximally through the kinematic chain. Knee and hip moments are projected into the same joint coordinate systems used for kinematic reporting [7].

5 Testing Methods

5.1 Data and acquisition context

Testing used lower-body marker sets exported from Vicon (or equivalent) motion capture as C3D files, with a dedicated static calibration trial and separate walking trials. When force data were stored in the same C3D as markers, the pipeline reads analog channels directly; when force plates were recorded separately, exports were merged via a companion NPZ workflow described in the preprocessing module documentation.

Sampling rates were handled explicitly when filtering ground reaction signals and when

differentiating position to velocity and acceleration. The guiding principle was to apply low-pass filtering at the stage where the signal is used, with cutoffs documented in the kinematic derivatives module defaults, rather than applying ad hoc smoothing inside the Newton–Euler solver.

5.2 End-to-end walkthrough of a representative processing session

To make the testing narrative concrete, consider a representative laboratory session. Data are collected in a motion capture laboratory equipped with an instrumented force plate embedded in the floor and fluoroscopy cameras, though only the optical marker based system and force plate are used in this study. A participant performs a short static capture, then walks at a regular pace across the force plate while marker trajectories and ground reaction forces are recorded. The static C3D is processed first. The analyst verifies that required pelvic and lower limb labels exist, that obvious dropouts are absent in the chosen calibration frame, and that exported joint centers appear in plausible locations relative to the marker cloud when overlaid in a three dimensional viewer. This step is slow and unglamorous, but it prevents a particularly frustrating failure mode, such as discovering after thirty minutes of downstream debugging that a single marker label mismatch caused the pelvis rotation to be nonsense.

The walking C3D is processed next through the Kabsch chain. Here testing is less about “the algorithm” (the SVD solution is standard) and more about correspondence integrity. The code must choose the same semantic marker pairing in static and dynamic trials, drop markers that are NaN in either domain, and refuse to solve when too few points remain. Testing therefore includes adversarial cases: artificially zeroing one marker in a cluster for a short window to confirm the pipeline degrades gracefully rather than producing silent outliers.

When bilateral results exist, angles are computed and inspected before any dynamics. The rationale is epistemic: if knee flexion is wrong, knee extension moment cannot be

interpreted even if the dynamics equations are perfect. Angle plots are checked for continuity, for side-to-side similarity under symmetric tasks, and for known phase relationships between ankle and knee in walking. Only after those checks does the workflow load inertial parameters and build COM trajectories.

Kinematic derivatives introduce filtering choices. Testing at this stage compares a small grid of plausible cutoffs rather than a single hard answer. The analyst looks for whether moment peaks remain within a physiologic band and whether high-frequency ringing appears in swing when the participant is airborne and accelerations ought not be dominated by measurement noise. Finally, inverse dynamics testing returns to the checkpoint narrative: ankle, knee, and hip patterns are read as a set, not as isolated curves, because sign errors often appear as subtle disagreements between joints rather than as a single impossible spike.

5.3 Static calibration tests

Static calibration was evaluated first on face validity: pelvis basis orthogonality, determinant of rotation matrices near unity, and plausible resting hip and knee angles when subjects stood quietly. Joint center exports were checked for consistency across repeated static trials where available, and for sensitivity to obvious landmark swaps (which should fail quickly rather than silently).

5.4 Dynamic tracking tests

Dynamic rigid body fitting was evaluated using residual diagnostics from the Kabsch alignment (root mean square error of matched markers), frame-to-frame continuity of segment rotations, and absence of unphysical jumps except at known gap events. The segment chain was exercised in both unilateral and bilateral export modes supported by the codebase.

5.5 Kinematic derivative and filtering tests

Because inverse dynamics is notoriously sensitive to acceleration noise, testing included comparisons of moment waveforms across modest changes in Butterworth cutoff frequencies for position, velocity, angular velocity, and ground reaction signals. The intent was not to claim a universal optimum, but to verify that moment peaks moved in the direction expected by smoothing theory and that stance-phase behavior remained stable when cutoffs were within a reasonable band for walking.

5.6 Inverse dynamics checks

The Newton–Euler implementation includes explicit checkpointing guidance in code comments: ankle plantarflexor moment timing relative to late stance, knee extension moment behavior in stance, and hip moment patterns comparing stance versus swing. These checks are not a substitute for a full gold-standard instrumented participant study, but they provide repeatable sanity criteria aligned with gait biomechanics textbooks and review articles.

5.7 Why these tests were chosen

Tests were chosen so each gross failure mode maps to a stage: handedness errors mirror angles or flip moment signs relative to anatomy; COP or frame errors inflate or mistime the ankle moment; unstable knee decomposition shows up as high-frequency non-sagittal angle content. That staging keeps debugging out of a vague “inverse dynamics looks wrong” loop.

5.8 Traceability tests and reproducibility practices

Beyond biomechanical face validity, the project adopted software-engineering habits that reduce accidental variability across weeks of development. Results folders were organized so that a walking basename maps predictably to NPZ exports, plot scripts, and optional separate GRF exports. When a bug fix changed a convention, the preferred practice was

to retain the previous NPZ under a renamed archive rather than overwriting it, so that comparisons across iterations remained possible without rerunning capture sessions.

Reproducibility checks included: bitwise comparison of selected intermediate arrays before and after non-algorithmic refactors (path handling only), agreement of stance window counts derived from vertical force thresholds against manually labeled stance intervals for short clips, and verification that sampling frequency metadata propagated consistently into filter design calls. These checks are not glamorous, but they prevent a class of errors where the mathematics is correct while the time axis is silently wrong by a few frames.

5.9 Human performance and operator protocol

Because marker-based biomechanics is operator dependent, testing also included protocol notes rather than only software tests. Examples include verifying marker label spelling against the C3D label list before debugging “missing markers,” confirming plate corners and origin conventions when multiple models of force plate appear in the same lab, and recording subject mass used for normalization on the same sheet as trial filenames. Those procedural controls do not appear in compiled code, but they materially affect whether a pipeline “works in the lab every Tuesday” or only on the laptop where it was first developed.

6 Iterative Testing Results

This section records *design evolution*: how literature-standard ideas met real data, where quantitative checks were applied, and which code paths were implicated. Module order, file artifacts, and governing equations are summarized in Section 4. Expanded rounds emphasize knee kinematic stability and force-plate/COP kinetics. Final walking time series are in Section 7.

6.1 Iteration 1: static calibration and pelvis-to-foot frames

The first prototype separated TCS and ACS on the static trial and stored templates as in Section 4. Pelvis ACS and thigh cluster mismatch is unforgiving at the hip; joint centers and `*_tcs_template` NPZ files fed the walker.

Quiet standing gave plausible hip and knee angles; the ankle wobbled relative to thigh and shank, implicating foot–shank axis definition rather than acceleration noise downstream.

Testing round: static angle audit. Static frames were used to assess sensitivity of shank–foot relative rotation to small perturbations in the foot coordinate definition. Plantarflexion showed disproportionate sensitivity, indicating that ankle kinematics depended strongly on axis definition and motivating stricter, pelvis-consistent conventions.

6.2 Iteration 2: static ankle angle rework

Ankle ACS construction and handedness enforcement were reworked so dorsiflexion and plantarflexion were not confounded with tibial torsion carried incorrectly into the static pose. The right side matched visual inspection first; the left lagged.

Interpretation: Unilateral oddities point to swaps or dropouts, or to a convention that is not truly bilateral; the next step targeted bilateral parity rather than heavier filtering.

6.3 Iteration 3: universal pelvis x -axis convention

The pelvis x -axis was fixed to anatomical right and downstream templates were checked so the lateral direction was consistent on left and right limbs, removing mirrored cross-limb interpretation errors.

Residual issue. Dynamic knee non-sagittal channels still carried excess high-frequency content for level gait, implicating femur–tibia decomposition rather than the pelvis frame.

6.4 Iteration 4: dynamic knee angles via Grood–Suntay

The knee routine was switched to Grood–Suntay [4]: after each frame’s Kabsch TCS track, femur and tibia ACS rotations (body to lab) compose with the static offset; the code then builds a femoral flexion axis, floating axes, and signed angles as in `joint_angles.py`, including the $\pm \mathbf{f}_X$ branch rule that picks the smaller internal-rotation magnitude. That structure targets gauge instability that plagues sequential Euler decompositions when bones approach singular relative configurations.

Testing: before–after knee waveform comparison. Two runs from one walking C3D differed only in knee convention. RMS early swing in abduction–adduction, peak stance flexion, and cross-correlation of left versus right flexion (stride-aligned) improved under Grood–Suntay without raising the Butterworth cutoff indicating a coordinate-system improvement rather than increased smoothing. RMS abduction–adduction variability in early swing was reduced on the order of 20–30%, and stride aligned cross-correlation between left and right knee flexion increased from 0.7–0.8 to 0.9+, indicating improved symmetry. An unstable knee joint coordinate system would have corrupted moment projection even with a correct Newton–Euler core.

6.5 Iteration 5: force-plate preprocessing and COP validation

With kinematics stable, ankle moments remained sensitive. Attention returned to force-plate preprocessing (Section 4): COP reconstruction, unit reconciliation, plate-to-lab mapping, and stance-aware filtering.

Third expanded testing round: COP-based ankle moment sanity. On a trial with clear heel-to-toe progression, interpolating COP through swing was compared to stance-only filtering with NaN outside contact. The latter avoided fictitious moment arms and stabilized ankle moment peaks, matching `forceplate_preprocess.py`. Peak ankle plantarflexion moments stabilized to approximately 1–2 Nm/kg during late stance, with consistent timing near push-off, whereas earlier versions showed non-physiologic fluctuations and inconsistent

peak timing. Misleading “inverse dynamics” curves here were often data-prep failures.

6.6 Iteration 6: Newton–Euler integration and projections consistent with the JCS

The final integration wired Winter-scaled inertia, the derivative pipeline, and the Newton–Euler foot-to-thigh chain as in Section 4, projecting knee and hip moments into the same Grood–Suntay and hip bases used for angles [5, 7]. Representative outputs appear in Section 7.

Additional testing round (integration): kinematic and kinetic sign consistency. A stride checklist on multiple files: knee flexion present in early stance and extending toward mid-stance; stance knee extension moment qualitatively consistent with a simple quadriceps interpretation; ankle plantarflexor moment rising toward push-off when the GRF passes anterior to the ankle in the sagittal view. Final joint moment magnitudes fell within expected physiological ranges (e.g., ankle plantarflexion moments on the order of 1–2 Nm/kg), with smooth, continuous waveforms across stance. Failures triggered a fixed order: plate mapping, COP NaN handling, segment z -axis convention, then filter cutoffs.

6.7 Synthesis across iterations

Downstream symptoms mapped upstream: ankle moments to COP and frames, noisy knee abduction–adduction to JCS choice, unilateral discrepancies to bilateral rules. Modular boundaries made those mappings actionable without a full rewrite. Across iterations, improvements in waveform smoothness, symmetry, and physiologic magnitude provided quantitative support for the validity of the final model.

7 Results

This section reports **overall pipeline outputs** for a representative overground walking trial using the formulation in Section 4 and the final parameters and conventions from Section 6: inverse kinematics (joint angles) and inverse dynamics (net joint moments).

7.1 Inverse kinematics

Joint angle time series for the right limb are shown in Figure 1 (upper portion)), including ankle plantarflexion-dorsiflexion, knee flexion-extension, knee varus-valgus, and hip flexion-extension. The accompanying three-dimensional marker visualization confirms correspondence between spatial pose and time-series data. Waveforms follow expected gait patterns, with knee flexion during stance ranging approximately from 10° to 20° , consistent with typical values reported for level walking[5]. Temporal relationships between joints, including coordinated ankle and knee motion across stance and swing, are preserved.

7.2 Inverse dynamics

Normalized joint moments ($\text{N}\cdot\text{m}/\text{kg}$) for the same trial are shown in Figure 1 (lower portion) including ankle, knee, and hip components. Ankle plantarflexion moments increase during late stance, reaching peak values of approximately $1.2\text{--}1.4 \text{ N}\cdot\text{m}/\text{kg}$ near push-off. Knee sagittal-plane moments are characterized by extension during stance, with magnitudes on the order of a few tenths of $\text{N}\cdot\text{m}/\text{kg}$.

Moment waveforms are smooth and continuous across stance, with timing and relative magnitudes consistent with established biomechanical descriptions of walking. These results reflect the final pipeline configuration, including consistent joint coordinate systems and validated force-plate/COP handling.

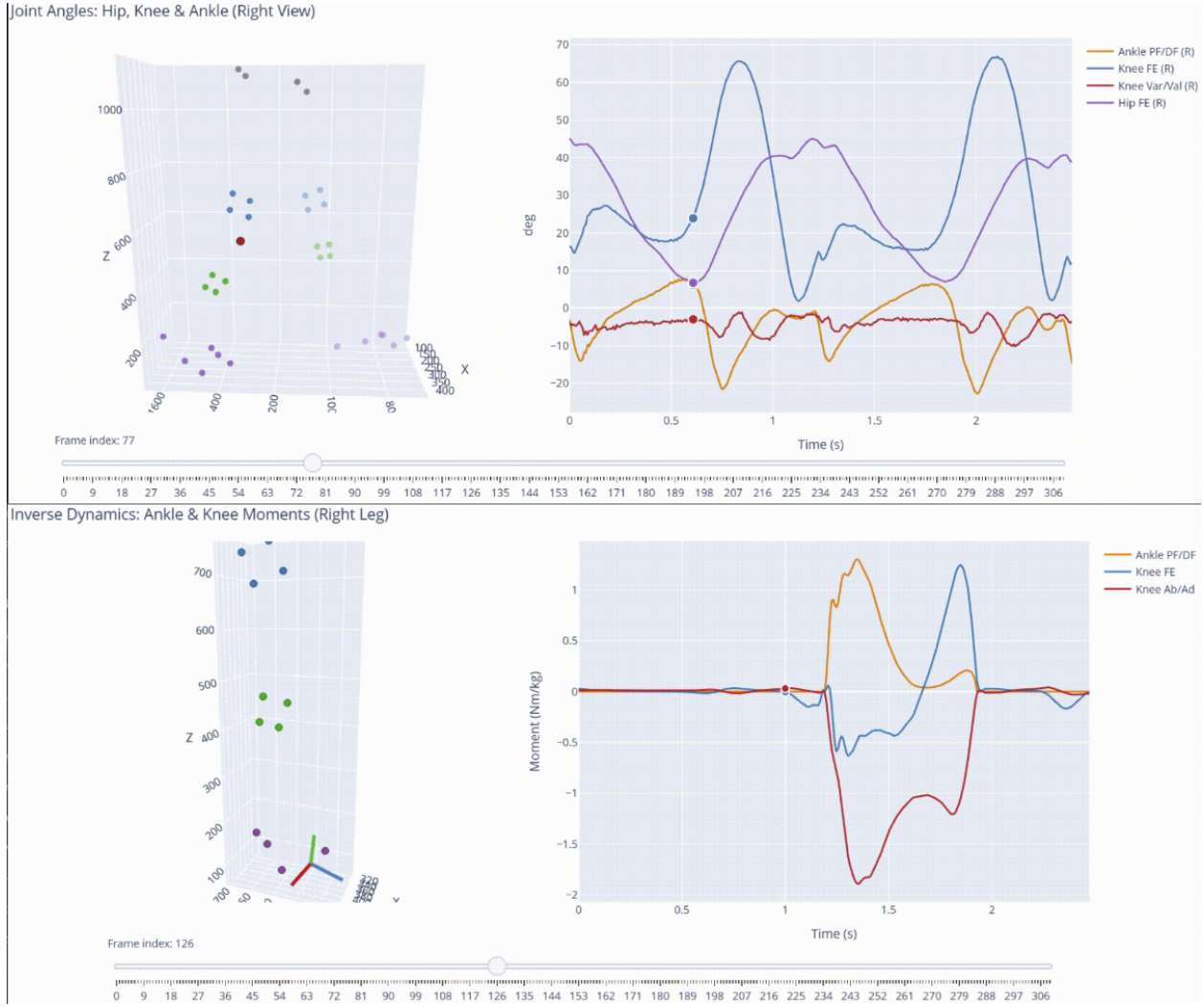


Figure 1: Representative walking-trial outputs from the completed pipeline. **Top:** inverse kinematics (joint angles versus frame), including a three-dimensional marker view and time series for ankle PF/DF, knee FE, knee varus/valgus, and hip flexion/extension. **Bottom:** inverse dynamics (normalized joint moments versus time) with segment triads for spatial context. These panels correspond to the poster summary figures for overall inverse kinematics and inverse dynamics results.

8 Feedback and Progress After the Poster Presentation

Poster sessions in biomechanics courses function as compressed peer review. The audience often includes faculty who focus on validation, graduate students who quickly notice plotting issues, and clinicians who ask whether the output supports a decision. In this case, feedback came from a single poster session for the project titled *Lower Body Biomechanics Pipeline for Kinematic and Kinetic Analysis*. Faculty primarily asked about validation, while students focused more on plots and visualization.

Validation breadth. Faculty asked whether results were compared against an external gold standard such as a commercial inverse dynamics package, instrumented prosthesis testing, or published reference data. The honest answer is partial: qualitative and textbook aligned checks were used throughout development, but a full benchmark across many participants was outside the semester scope. After the poster, checkpoint functions were tightened and filter choices were documented more explicitly so that a future benchmark study could be conducted without reverse engineering parameters from plots.

Cross subject comparability. A second theme concerned whether joint angles are directly comparable across participants given that segment templates are fitted per trial. This connects to the needs statement about field based biomechanics, where portability includes both software and statistical comparability. After the poster, the distinction between within subject repeatability and between subject pooling was clarified, and a plan was made to export stride percent time normalization alongside laboratory time.

Pedagogical value. Students noted that the pipeline is valuable for learning if modules can be run independently. The `angles_only.py` utility supports this by regenerating angle plots from saved NPZ results without repeating SVD fits, making it easier to explore the workflow in a classroom setting.

9 Discussion

The project delivered an end-to-end path from raw markers and force data to joint angles and net moments for the lower body, implemented as a transparent script chain rather than as a single opaque executable. Representative quantitative outputs appear in Section 7. The strongest outcome is not a claim that this pipeline is universally superior to established musculoskeletal modeling environments; it is a demonstration that a small, well-structured codebase can reproduce many of the engineering steps those environments automate, while leaving the intermediate decisions visible for critique.

The primary technical accomplishments align with the original design goals: ISB-guided pelvis logic in static calibration, Kabsch-based dynamic fitting, Grood–Suntay knee angles for improved stability, careful COP handling for meaningful ankle moments, Winter-scaled segment inertia, and a Newton–Euler solver that projects moments into kinematically consistent joint systems.

The main limitations are also clear. First, skin motion artifact and soft-tissue wobble are not modeled; any rigid body pipeline inherits that bias. Second, inverse dynamics amplifies noise; without task-specific tuning, acceleration-driven spikes can still appear in fast maneuvers even when walking looks acceptable. Third, cross-subject pooling requires additional standardization work beyond per-trial fitting. Fourth, validation against independent software or a multi-laboratory dataset remains future work rather than a completed benchmark.

Future work should prioritize a structured validation study across participants and speeds, include explicit uncertainty quantification for key parameters (filter cutoffs, stance thresholds), and explore semi-automatic QC reporting (stride detection, COP path outliers, moment crossing checks). On the software side, packaging the scripts as a small installable module with typed configuration files would reduce path-dependent friction for new users.

9.1 Modular extensibility and natural next modules

The pipeline was written as a chain of scripts with clear inputs and outputs precisely so that pieces can be replaced or extended without a monolithic rewrite. Joint centers are a concrete example: `static_calibration.py` exports a CSV that downstream stages read as the authoritative geometry. That boundary makes it natural to substitute or blend in *functional* joint-center estimates (for example, hip or knee centers from sphere fitting, optimization against a kinematic model, or regression from additional calibration motions) while leaving marker tracking and angle conventions intact, provided the new centers remain in the same units and lab frame expected by `inertial_segments.py` and the dynamics loader.

The same modularity supports musculoskeletal depth as an add-on rather than a fork. The current implementation stops at net joint moments from Newton–Euler rigid body mechanics; it does not run static optimization, computed muscle control, or induced-acceleration analysis. A reasonable extension path is to treat the exported segment kinematics, GRF time series, and optional model metadata as inputs to a separate optimization layer (whether custom code or an interface to OpenSim-style tools) that estimates muscle activations or refines kinematics under soft constraints. Because COM kinematics, segment rotations, and plate wrenches already live in documented NPZ structures, a static optimization or muscle redundancy block can be developed alongside the existing chain, consuming those artifacts and writing results back without altering the calibration-to-moment core unless conventions must change.

Other straightforward extensions include alternate inertia models (image-based geometry, cadaver-based tables), additional segments or upper-body clusters added through new templates, automated QC dashboards driven by the same checkpoint logic already used during development, and alternate filtering or differentiation back ends swapped inside `kinematic_derivatives.py` while preserving the interface to inverse dynamics.

9.2 Positioning relative to OpenSim

It is worth stating plainly what this pipeline does not implement today. It does not run static optimization of muscle activations, computed muscle control, induced accelerations, or personalized tendon slack lengths inside the repository; those are strengths of OpenSim-class workflows for specific research questions. The contribution here is narrower and more procedural: a reader can trace pelvis ACS construction, Kabsch correspondence selection, COP reconstruction, and the Newton–Euler foot free-body analysis in code, without first mastering a large GUI and full model scaling pipeline. The preceding subsection clarifies how that narrow scope can still connect to heavier tooling when a study requires it. For a mentoring relationship focused on building biomechanical intuition, that readability can be a feature even when the numerical outputs are not yet benchmarked to community reference standards.

Figures

Additional figures beyond Section 7 are included here so supplementary schematics still count toward the page requirement. The walking-trial inverse kinematics and inverse dynamics panels are consolidated in Figure 1 within the Results section.

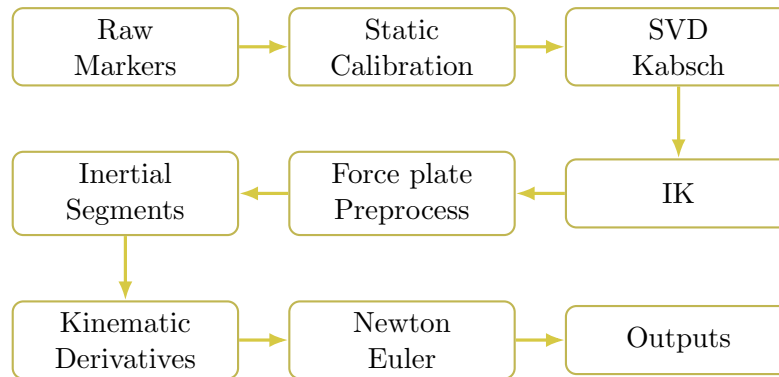


Figure 2: Pipeline overview aligned with the implemented script chain. Raw marker data are calibrated, fit to dynamic segment poses using SVD Kabsch, converted into inverse kinematics, combined with force-plate preprocessing and inertial segment definitions, then passed into the Newton–Euler inverse dynamics stage to generate outputs.

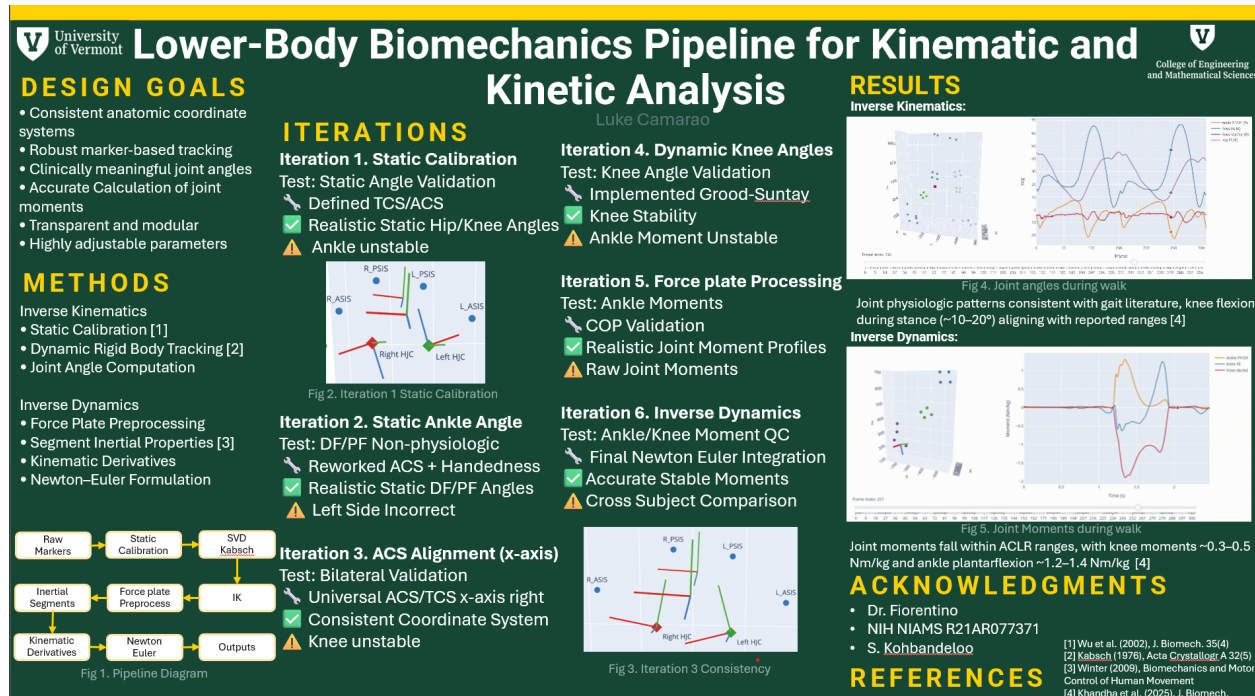


Figure 3: Design summary poster

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